ASSIGNMENT-3

PCCST501 – COMPUTER NETWORKS

Assignment 3 | Module 3 – Data-Link Layer, Wireless LANs & Mobile IP

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1. Explain the responsibilities of the data-link layer and distinguish logical link control from media access control.

A. Responsibilities of the Data-Link Layer

The data-link layer is the lowest layer of the TCP/IP protocol stack (also called the link layer or network access layer). It is responsible for transmitting data between two directly connected devices over a physical network. Communication at this layer is node-to-node (hop-to-hop): data travels from one node to the next across one link at a time, through the source host, routers and destination host (Figure 1).

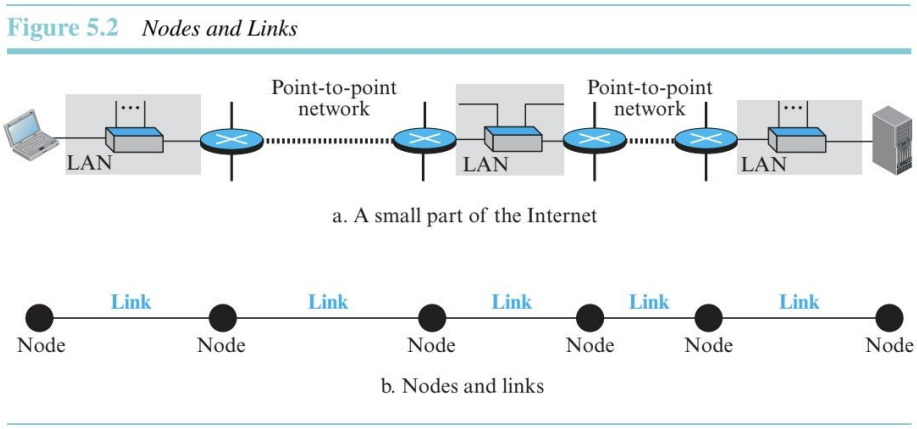


Figure 1. Nodes and links. Data moves one link at a time from node to node (course module, slide 3).

Its main responsibilities are:

1. **Framing:** The bit stream from the physical layer is grouped into frames. Each frame gets a header (control information including the source and destination MAC addresses) and a trailer (error-detection bits, CRC).
2. **Physical (MAC) addressing:** Source and destination link-layer addresses are inserted into the frame. These addresses change at every link, while the IP addresses stay the same from source to destination.
3. **Flow control:** The rate of frame transmission between two adjacent nodes is coordinated so that the receiver is not overwhelmed and its buffers do not overflow.

4. **Error control:** Damaged or lost frames are detected and, where the protocol supports it, retransmitted on the link. Most wired link-layer protocols discard corrupted frames and leave recovery to higher layers.
5. **Access control:** When many devices share one medium (a broadcast link), a multiple-access method decides which device may transmit at a given time so that frames do not collide.

B. Two Sublayers of the Data-Link Layer (Course View)

The course module divides the data-link layer into two sublayers. The **Data Link Control (DLC)** sublayer handles functions common to point-to-point and broadcast links, namely framing, flow control and error control. The **Media Access Control (MAC)** sublayer handles functions that are specific to shared (broadcast) links: it controls how multiple devices access the common medium and uses physical (MAC) addresses. Figure 2 shows that a point-to-point link needs only the DLC sublayer, whereas a broadcast link needs both.

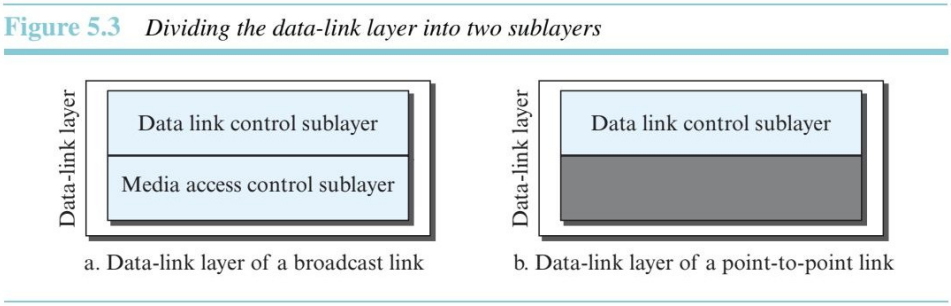


Figure 2. Dividing the data-link layer into two sublayers (course module, slide 5).

C. Difference Between Logical Link Control (LLC) and Media Access Control (MAC)

In the IEEE 802 LAN standards, the data-link layer is divided into LLC and MAC sublayers. The distinction is summarised in the table below.

Aspect	Logical Link Control (LLC)	Media Access Control (MAC)
Position	Upper sublayer of the data-link layer	Lower sublayer of the data-link layer
Interface	Provides an interface to the network layer	Provides an interface to the physical layer
Main purpose	Gives one common link-layer control protocol for all IEEE LANs; identifies the network-layer protocol carried in the frame	Decides how devices access and share the network medium; handles MAC addressing and part of framing
Functions	Flow control, error control, part of framing	Defines the access method for each LAN (for example CSMA/CD for Ethernet, CSMA/CA for Wi-Fi)
Effect on upper layers	Makes the MAC sublayer transparent to the network layer	Hides the details of the shared medium from LLC
Example standard	IEEE 802.2 LLC	Ethernet MAC (IEEE 802.3) and Wi-Fi MAC (IEEE 802.11)

Conclusion: LLC is concerned with the logical link between two protocol entities (for example, IP running above a LAN), whereas MAC is concerned with the physical access to the shared medium.

Because of LLC, the same network-layer protocol can run over Ethernet, Token Ring or Wi-Fi without change.

2. Explain Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.

A. Ethernet Frame Format

Ethernet is the most popular LAN technology. It was developed in the 1970s, uses CSMA/CD for access control on the original shared medium, and operates at the data-link layer. Data is carried in Ethernet frames, which have the structure shown in Figure 3 and the table below.

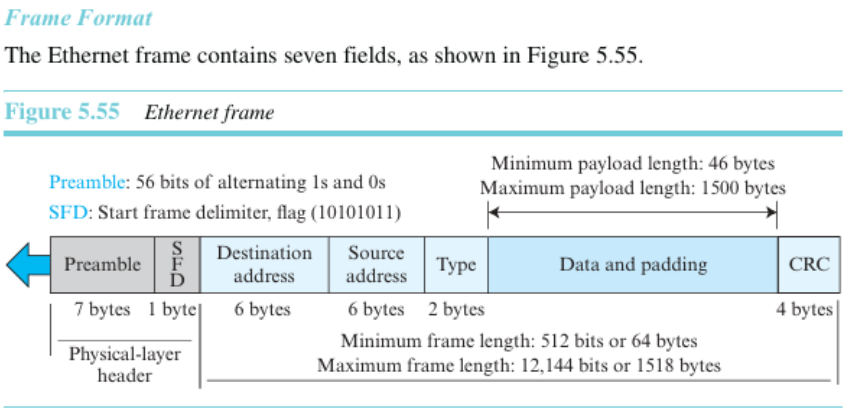


Figure 3. Ethernet frame format with the field sizes (course module, slide 118, Figure 5.55).

Field	Size	Purpose
Preamble	7 bytes	Alternating 1s and 0s that let the receiver synchronise its clock (added at the physical layer)
SFD (Start Frame Delimiter)	1 byte (10101011)	Marks the beginning of the frame and gives the receiver a final synchronisation chance
Destination address (DA)	6 bytes	MAC address of the intended receiver (unicast, multicast or broadcast)
Source address (SA)	6 bytes	MAC address of the sender
Type / Length	2 bytes	Identifies the upper-layer protocol (Type) or gives the data length (Length)
Data and padding	46 to 1500 bytes	Payload from the upper layer; padded to 46 bytes when shorter
FCS (CRC-32)	4 bytes	Frame Check Sequence used to detect transmission errors

B. Purpose of Each Field

1. **Destination MAC address (6 bytes):** Identifies the device on the local network that should receive the frame. If the first bit of the first byte is 0 the address is unicast; if it is 1 the address is multicast. The all-ones address FF:FF:FF:FF:FF:FF is the broadcast address, which is received by all stations on the LAN.

2. **Source MAC address (6 bytes):** Identifies the device that sent the frame. A source address is always unicast, because one device sends a frame. Switches use it to learn which port a station is on (see Question 4).
3. **Type/Length field (2 bytes):** Its meaning depends on the value. If the value is 0x0600 or greater, it is an EtherType that identifies the upper-layer protocol, for example 0x0800 for IPv4 and 0x0806 for ARP, which helps the receiver pass the data to the correct protocol (multiplexing and demultiplexing). If the value is 1500 or less, it gives the length of the data field, as in IEEE 802.3.
4. **Frame Check Sequence (FCS, 4 bytes):** Contains a 32-bit CRC value computed over the frame. The receiver recalculates the CRC; if it does not match, the frame is discarded. Ethernet does not retransmit it, so recovery is left to higher layers such as TCP.

C. Other Fields and Frame Limits

- **Preamble (7 bytes) and SFD (1 byte):** Added by the physical layer and normally not counted as part of the frame.
- **Data and padding (46 to 1500 bytes):** If the data is shorter than 46 bytes, padding is added. Data longer than 1500 bytes must be split into several frames.
- **Frame size:** The minimum frame is 64 bytes (512 bits) and the maximum is 1518 bytes. Header plus trailer is 18 bytes (14 bytes of header and 4 bytes of FCS). The 64-byte minimum is needed for CSMA/CD to work (Question 6).
- **Service:** Ethernet is connectionless and unreliable. It does not set up a connection, and it discards corrupted frames without retransmitting them.

3. Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.

Aim: To capture Ethernet traffic with Wireshark and identify the source and destination MAC addresses of at least five frames.

Software required: Wireshark and a computer with a network connection.

A. Procedure

1. Open Wireshark and select the active network interface (Ethernet or Wi-Fi).
2. Click Start Capturing Packets.
3. Generate traffic, for example by opening a web page or running ping to another device on the LAN.
4. Stop the capture after enough packets have been recorded.
5. Apply the display filter **eth** to show Ethernet frames.
6. Select any five packets and expand the **Ethernet II** section in the packet details pane.
7. Record the Source and Destination MAC addresses shown for each frame.

B. Observation Table

Note on the data: *The values below are* ***illustrative sample values*** *arranged to show the typical patterns (unicast, reply, broadcast and multicast). They are not the result of a real capture made for this report. When the assignment is done on your own machine, replace them with the values from your own Wireshark capture and attach a readable screenshot of the Ethernet II section.*

Frame No.	Source MAC address	Destination MAC address	Type of frame (from destination)
1 (sample)	3C:52:82:A1:0F:4B	FF:FF:FF:FF:FF:FF	Broadcast (for example, an ARP request)
2 (sample)	00:1E:C9:5A:7B:2D	3C:52:82:A1:0F:4B	Unicast (reply sent to the PC)
3 (sample)	3C:52:82:A1:0F:4B	00:1E:C9:5A:7B:2D	Unicast (PC sends to the default gateway)
4 (sample)	00:1E:C9:5A:7B:2D	3C:52:82:A1:0F:4B	Unicast (gateway reply)
5 (sample)	3C:52:82:A1:0F:4B	01:00:5E:00:00:FB	Multicast (group address)

C. How the Addresses Were Identified

- The first transmitted bit of the first byte of the destination address decides between unicast and group addresses: 0 means unicast and 1 means multicast or broadcast. Frame 1 and frame 5 show this.
- The source address is always unicast, because a frame has only one sender.
- Address bytes are transmitted left to right, and within each byte the least significant bit is sent first (course module, slides 123 to 125).

Result: The source and destination MAC addresses of five Ethernet II frames were identified from the packet details pane using the procedure above. The address patterns shown above are those expected from a real capture.

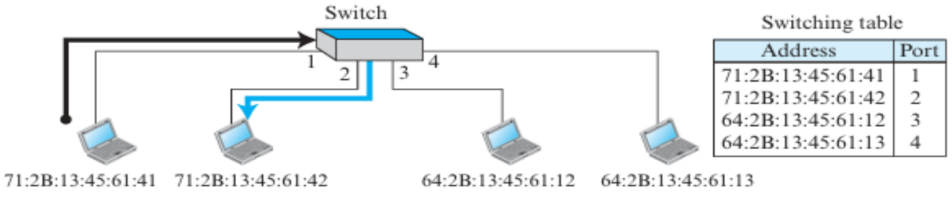
4. Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.

A. Introduction

A link-layer switch connects several stations on a LAN and forwards each Ethernet frame only to the port where its destination is. It works at Layer 1, where it regenerates the signal, and at Layer 2, where it reads MAC addresses. Unlike a hub, it does not send every frame to every port. It learns the location of each station automatically and stores this in a **forwarding table** (also called a MAC address table or switching table). Figure 4 shows the idea.

Let us give an example. In Figure 5.85, we have a LAN with four stations that are connected to a link-layer switch. If a frame destined for station 71:2B:13:45:61:42 arrives at port 1, the link-layer switch consults its table to find the departing port.

Figure 5.85 *Link-Layer Switch*



According to its table, frames for 71:2B:13:45:61:42 should be sent out only through port 2; therefore, there is no need for forwarding the frame through other ports.

A link-layer switch does not change the link-layer (MAC) addresses in a frame.

Figure 4. A link-layer switch forwards the frame only to the port of the destination (course module, slide 216, Figure 5.85).

B. Operation of a Switch Using MAC Address Learning

For every frame it receives, the switch performs the following steps:

1. **Frame reception:** The frame arrives on an input port.
2. **Source MAC learning:** The switch reads the source address and records it with the incoming port, in the form source address -> port. The learning is done on the source address, not the destination.
3. **Destination MAC lookup:** The switch searches the table for the destination address.
4. **Forwarding:** If the destination is known and is on a different port, the frame is sent only to that port.
5. **Flooding:** If the destination is unknown, the frame is sent to all ports except the incoming port.
6. **Filtering:** If the destination is on the same port the frame arrived on, the frame is not forwarded at all.
7. **Table updating and aging:** If a station's address is seen on a different port, its entry is updated. Entries that are not refreshed are removed when their aging time expires.

C. Demonstration: How a Switch Builds Its Forwarding Table

Consider four computers connected to a four-port switch: Computer A on Port 1, Computer B on Port 2, Computer C on Port 3 and Computer D on Port 4. The switch's table starts empty (Figure 5 shows the same learning process in the course module).

Step	Frame (source -> destination)	Learned from source	Destination known?	Action taken	Table after the step
1	A -> D (arrives on Port 1)	A is on Port 1	No (D unknown)	Flooded to Ports 2, 3 and 4	A - 1
2	D -> B (arrives on Port 4)	D is on Port 4	No (B unknown)	Flooded to Ports 1, 2 and 3	A - 1, D - 4
3	B -> A (arrives on Port 2)	B is on Port 2	Yes (A is on Port 1)	Forwarded only to Port 1	A - 1, B - 2, D - 4
4	C -> D (arrives on Port 3)	C is on Port 3	Yes (D is on Port 4)	Forwarded only to Port 4	A - 1, B - 2, C - 3, D - 4

Explanation of the steps: In step 1 the switch learns where A is, but it does not yet know D, so it floods the frame. In step 2 it learns D, and because B is still unknown it floods again. From step 3 onwards the switch knows both source and destination, so it forwards each frame only through the correct port (filtering). After step 4 all four stations are learned.

Final forwarding table:

MAC address	Switch port
A	1
B	2
C	3
D	4

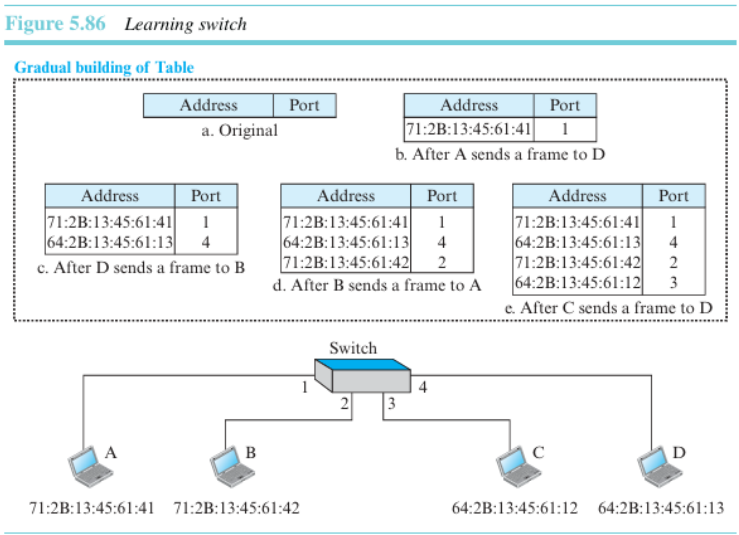


Figure 5. Learning switch: how the table is built up frame by frame (course module, slide 218, Figure 5.86).

Points to note: A station that never sends a frame is never learned, so frames to it keep being flooded. Learning also takes time, which is why the first frame to a new station is usually flooded. Loops are prevented by the spanning tree protocol (IEEE 802.1D), which is part of transparent switching.

5. Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.

A hub, a switch and a bridge all connect devices in a network, but they differ in the layer at which they work, how they forward frames and how many collision domains they create.

A. Short Description of Each Device

- **Hub:** A multiport repeater working at the physical layer (Layer 1). It has no MAC table and no filtering capability; it regenerates the signal and sends each incoming frame out of all ports except the one it came in on.
- **Switch:** A link-layer device (Layer 2, with a physical-layer function). It reads the destination MAC address and forwards the frame only to the correct port, using a dynamically learned forwarding table.
- **Bridge:** A link-layer device that connects two or more LAN segments. Like a switch it learns MAC addresses and filters frames, but it usually has fewer ports, and each segment it connects is a separate collision domain. Early bridges were software based; a modern multiport bridge is essentially a switch.

B. Comparison Table

Feature	Hub	Switch	Bridge
OSI layer	Physical (Layer 1)	Data-link (Layer 2)	Data-link (Layer 2)
Forwarding method	Repeats the frame to all ports	Forwards to the destination port using the MAC table	Forwards frames between segments based on MAC addresses
Filtering	None	Yes, per frame	Yes, per frame
Collision domains	One shared collision domain for all ports	Each port is its own collision domain (full duplex: none)	Each connected segment is a separate collision domain
Broadcast domains	One	One (or one per VLAN when VLANs are configured)	One
Performance	Low (shared bandwidth)	High (dedicated bandwidth per port)	Moderate
Practical use	Simple, older LANs; now largely obsolete	Modern LANs, data centres, campus networks	Joining or segmenting LAN sections; the switch is now the more common choice

C. Forwarding Summary

The table below summarises the three forwarding behaviours. The hub repeats the frame to every station, the switch sends it only to the port of the destination, and the bridge connects two LAN segments and forwards only the frames that need to cross between them.

Device	What happens to a frame from PC A to PC B
Hub	PC A -> Hub -> every port (PC B and all other PCs receive the frame; only PC B keeps it)
Switch	PC A -> Switch -> Port of PC B only (other ports do not see the frame)
Bridge	LAN 1 -> Bridge -> LAN 2 (only frames addressed to stations on LAN 2 cross the bridge)

Conclusion: A hub creates one large collision domain, while a switch or bridge divides the LAN into separate collision domains. For this reason, switches have replaced hubs in modern networks.

6. Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.

A. Definition

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is a random-access medium access control method used in traditional shared, half-duplex Ethernet. Stations sense the medium before they transmit and continue to monitor it during transmission so that they can detect collisions.

B. Working of CSMA/CD

1. **Carrier sense:** A station listens to the channel before transmitting.
2. **Transmission:** If the channel is idle, the station starts transmitting immediately (1-persistent method).
3. **Collision detection:** While transmitting, the station keeps monitoring the medium. If it detects another signal, a collision has occurred.
4. **Abort and jam:** The station stops sending the frame and transmits a short jam signal so that all stations know about the collision.
5. **Binary exponential backoff:** After the n-th collision the station waits for a random number of slot times R, chosen from 0 to 2^n - 1, and then tries again. The range doubles after each collision, which reduces the chance of repeated collisions. After 16 failed attempts the frame is discarded.
6. **Success:** If no collision is detected for the whole frame, the transmission is complete.

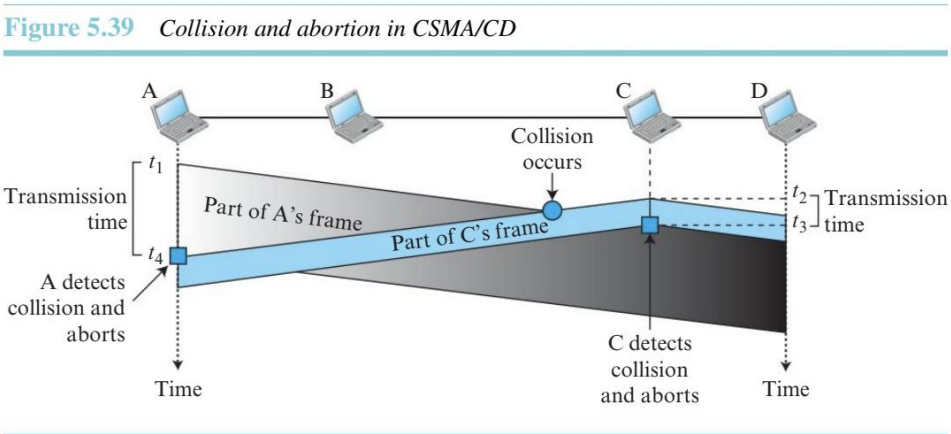


Figure 7. Collision and abortion in CSMA/CD: station C detects the collision and stops, while station A also detects it (course module, slide 87, Figure 5.38).

C. Minimum Frame Size and the 2Tp Rule

For collision detection to work, the sender must still be transmitting when the collision signal returns from the farthest station. The worst case is the signal travelling to the far end (Tp) and the collision information coming back (Tp), so the total time is 2Tp. The minimum frame transmission time must therefore be at least 2Tp: $T_{fr} \geq 2T_p$ (course module, slides 88 and 89). This is why Ethernet frames have a minimum size of 64 bytes (512 bits).

D. Why Its Importance Is Reduced in Modern Ethernet

- **Dedicated links:** In switched Ethernet each station has its own point-to-point link to the switch, so stations do not compete for a shared medium.
- **Full-duplex operation:** A station can send and receive at the same time on its own link. A collision is therefore not possible during normal operation, because frames do not share a channel.
- **Buffering:** The switch stores incoming frames in buffers and sends them out through the correct port, so it does not need to avoid collisions on the output.
- **Higher speeds:** Fast Ethernet with a switch and full duplex, Gigabit Ethernet and 10-Gigabit Ethernet use full-duplex links; the 10-Gigabit standard operates only in full-duplex mode, so CSMA/CD is not used in it (course module, slides 135, 137 and 140).
- **Limits set by other factors:** In switched full-duplex networks the maximum cable length is set by signal attenuation, not by collision detection.

Conclusion: CSMA/CD was essential in the original shared Ethernet, where many stations used one cable. In modern switched full-duplex Ethernet collisions do not occur in normal operation, so CSMA/CD is no longer needed. The protocol is still defined for half-duplex operation, such as in some Gigabit Ethernet implementations, but it is rarely used in practice.

7. Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.

A. Definition

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) is the access method used by the Distributed Coordination Function (DCF) in IEEE 802.11 wireless LANs (Wi-Fi). Its purpose is to **avoid** collisions before they happen, because a wireless station cannot reliably detect a collision while it is transmitting.

B. Working of CSMA/CA

CSMA/CA uses three main strategies: an interframe space, a contention window with random backoff and acknowledgements. The sequence is as follows:

1. **Sense the channel:** The station checks whether the wireless medium is idle. If it is busy, the station waits until it becomes idle.
2. **Wait an interframe space:** When the channel is idle, the station waits for DIFS (the distributed interframe space).
3. **Random backoff:** The station chooses a random number of time slots from the contention window and counts them down. The contention window uses binary exponential backoff: its size increases after each failed transmission. If the channel becomes busy during the countdown, the

timer pauses and resumes when the channel is idle again.

4. **Reservation (optional):** The station may send a Request to Send (RTS). The destination answers with Clear to Send (CTS). All other stations read the duration in these frames and set their Network Allocation Vector (NAV) timer, so they keep quiet for the whole transmission. This solves the hidden-station problem (Figure 9).
5. **Transmit the data frame:** The sender transmits the frame.
6. **Acknowledgement:** The receiver waits SIFS (a shorter interval than DIFS) and sends an ACK. If no ACK arrives before the timeout, the sender assumes the frame was lost, increases its retry count and tries again. After the retry limit is reached, the frame is dropped.

frames in time.

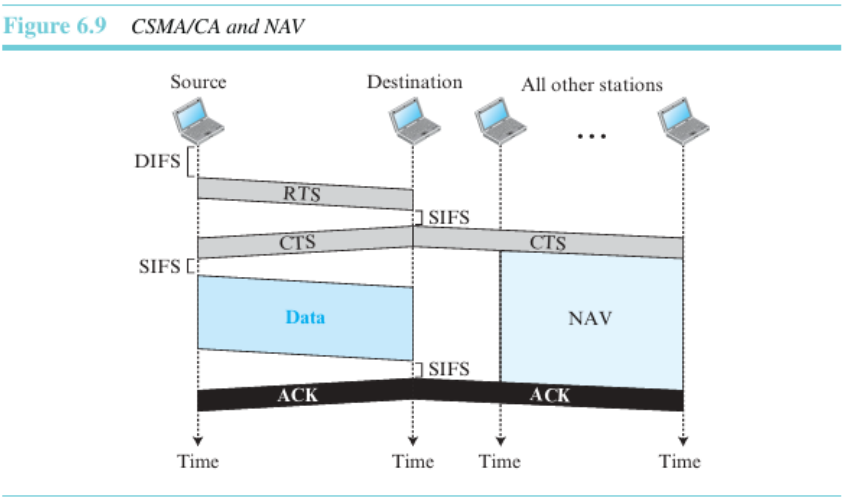


Figure 8. Frame exchange timeline with DIFS, SIFS, RTS, CTS, data, ACK and the NAV (course module, slide 162, Figure 6.9).

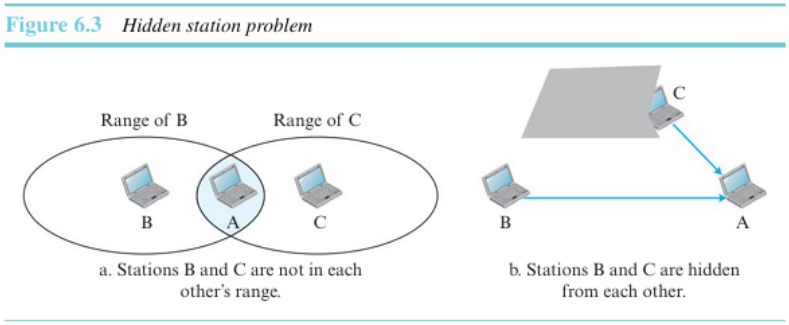


Figure 9. Hidden-station problem: B and C cannot hear each other but both reach A (course module, slide 153, Figure 6.3).

C. Comparison Between CSMA/CA and CSMA/CD

Feature	CSMA/CA (IEEE 802.11)	CSMA/CD (traditional Ethernet)
Full form	Carrier Sense Multiple Access with Collision Avoidance	Carrier Sense Multiple Access with Collision Detection
Main objective	Avoid collisions before transmission	Detect collisions during transmission and recover
Medium	Wireless (radio)	Wired (coaxial or twisted pair)
Can the station detect collisions?	No, a radio cannot transmit and listen at once; collisions are inferred from a missing ACK or CTS	Yes, the station monitors the cable while transmitting
Acknowledgement	Used (ACK from the receiver) because losses are common	Not part of the basic CSMA/CD method
Extra mechanisms	IFS, random backoff, RTS/CTS, NAV	Jam signal, binary exponential backoff, minimum frame size
Response to a collision	Frame not sent successfully is retried after backoff	Transmission stopped at once, jam sent, retry after backoff
Frame size constraint	No minimum-frame / 2Tp requirement	Minimum frame size of 64 bytes so 2Tp is covered

Conclusion: CSMA/CD tries to detect a collision after it starts, while CSMA/CA tries to prevent it from happening. CSMA/CA is used in Wi-Fi because wireless devices cannot listen while they transmit and because hidden and exposed stations cause problems that a simple collision-detection scheme cannot solve.

8. Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.

A. Aim

To design a secure and reliable wireless LAN (WLAN) for a college computer laboratory that gives students and staff access to the internet and to college resources.

Assumptions: The laboratory holds about 40 users at a time, with a mix of student laptops, staff laptops, mobile phones and about 30 wired lab PCs. The WLAN works in infrastructure mode: wireless hosts connect to an access point (AP), which is connected to the wired college network (course module, slide 150).

B. Network Diagram

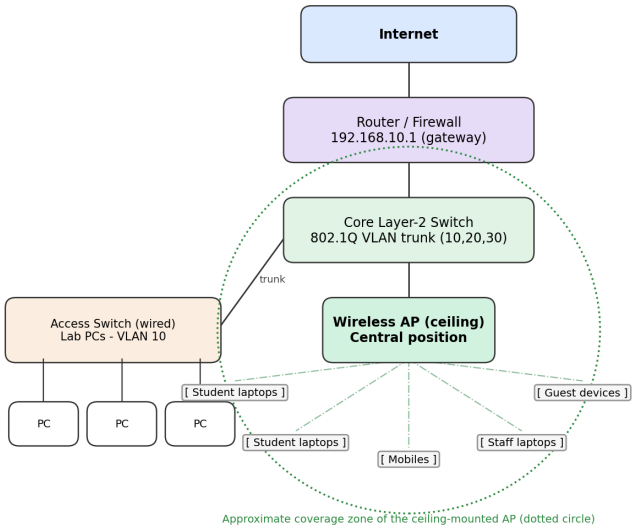


Figure 10. Proposed WLAN design for the college laboratory. This diagram was drawn for this report and is not taken from the course module.

C. Network Design Specifications

Parameter	Specification
Access point placement	One dual-band AP mounted on the ceiling in the centre of the laboratory, away from metal cabinets, thick walls and microwave ovens. A second AP is added if the room is large or if the seat count exceeds about 25 wireless users per AP. The placement aims for a clear line of sight to most seats.
SSID	Three SSIDs, one per user group, each mapped to its own VLAN: **CS-LAB-STU** (students), **CS-LAB-STAFF** (staff) and **CS-LAB-GUEST** (visitors). An SSID is the name that identifies the WLAN and is broadcast in beacon frames.
Frequency band	Dual-band operation: 5 GHz for higher speed in the room, and 2.4 GHz for range and older devices. Devices that use the 2.4 GHz band can be affected by Bluetooth, which also uses it (course module, slide 177).
Channel selection	2.4 GHz: use channels 1, 6 and 11 (20 MHz width), which do not overlap. 5 GHz: use non-overlapping channels such as 36, 40, 44 and 48, and avoid DFS channels where possible. Choose the channel after a site survey, because neighbouring APs may interfere.
IP addressing	Each VLAN has its own subnet: students 192.168.10.0/24, staff 192.168.20.0/24, guests 192.168.30.0/24 (see the addressing plan below). Default gateway 192.168.10.1 (router or firewall interface). Clients receive addresses by DHCP. The AP's management address is static.
DNS	College DNS server for staff and student networks; a trusted public resolver for the guest network.

D. Addressing Plan

Each VLAN is a separate broadcast domain, with a separate subnet and DHCP scope. The plan is set out below. Wireless clients keep the same IP address on the wired side because the AP bridges the wireless and wired environments. Only the link-layer (MAC) address changes at each hop.

VLAN	Users	Subnet	Gateway	DHCP range
10	Students (CS-LAB-STU) and lab PCs	192.168.10.0/24	192.168.10.1	192.168.10.100 to 192.168.10.200
20	Staff (CS-LAB-STAFF)	192.168.20.0/24	192.168.20.1	192.168.20.100 to 192.168.20.200
30	Guests (CS-LAB-GUEST)	192.168.30.0/24	192.168.30.1	192.168.30.100 to 192.168.30.200

E. Basic Security Requirements

1. **Strong encryption:** Use WPA3-Personal (SAE) for the student network and WPA3-Enterprise (802.1X with individual accounts) for staff. Do not use open networks or WEP.
2. **Separate networks:** Keep student, staff and guest traffic on separate VLANs carried by IEEE 802.1Q trunk links, so that broadcasts are limited to members of each VLAN (course module, slide 145).
3. **Firewall rules:** The guest VLAN may reach the internet but not the college servers or the other VLANs. Only the necessary ports are open to the lab PCs.
4. **Client isolation:** Enable client isolation on the guest SSID so that guest devices cannot reach each other.
5. **WPS and default settings:** Disable Wi-Fi Protected Setup and change the default administrator password of the AP and router.
6. **Management access:** Manage the AP only from the staff VLAN, over an encrypted connection, and restrict management to named administrators.
7. **Updates and logging:** Keep the AP and router firmware up to date and keep logs of association and administrator activity.

Conclusion: The design gives every user group a suitable SSID and VLAN, places the AP for good coverage of the laboratory, uses non-overlapping channels, and applies WPA3 and VLAN separation. Guest access is restricted so that it cannot reach college resources.

9. Explain Mobile IP and the need for mobility support when a device moves between networks.

A. Definition

Mobile IP is an extension of the IP protocol that allows a mobile device, such as a laptop, to stay connected to the internet while it moves between networks. It maintains continuous communication even when the device changes its network location, so the device can connect from any place where an internet connection is available.

B. Need for Mobility Support

IP addresses are designed for stationary hosts. An IP address has a network prefix, which tells routers where to deliver a packet, and a host suffix. If a host moves to another network, its address no longer matches the prefix of the network it is on. Routers would then send packets to the old network and they would never arrive. Simply giving the host a new address would also break existing TCP connections, because IP addresses and ports must stay constant during a connection. Mobile IP therefore provides:

- **Continuous connectivity:** communication continues while the device moves between networks.
- **A permanent identity:** the device keeps its home IP address, so remote hosts and applications need not change.
- **Location tracking:** the home network always knows where the device currently is.
- **Transparency:** the change of network is hidden from applications and from the correspondent host as far as possible.

C. Components of Mobile IP

Component	Role
Mobile node (MN)	The device that moves between networks
Home agent (HA)	A router on the home network that tracks the current location of the mobile node and tunnels packets to it
Foreign agent (FA)	A router on the visited network that receives tunnelled packets from the home agent and delivers them to the mobile node (Mobile IPv4)
Care-of address (CoA)	A temporary address that represents the mobile node's position on the foreign network
Correspondent node (CN)	The host that communicates with the mobile node

D. Two Addresses: Home Address and Care-of Address

The main solution is that the mobile host uses two addresses. The **home address** is permanent and belongs to the home network; it does not change when the host moves. The **care-of address** is temporary and belongs to the foreign network where the host currently is. Figure 11 shows the two addresses. Changing the address itself (using DHCP on each move) was rejected because it interrupts connections and requires reconfiguration.

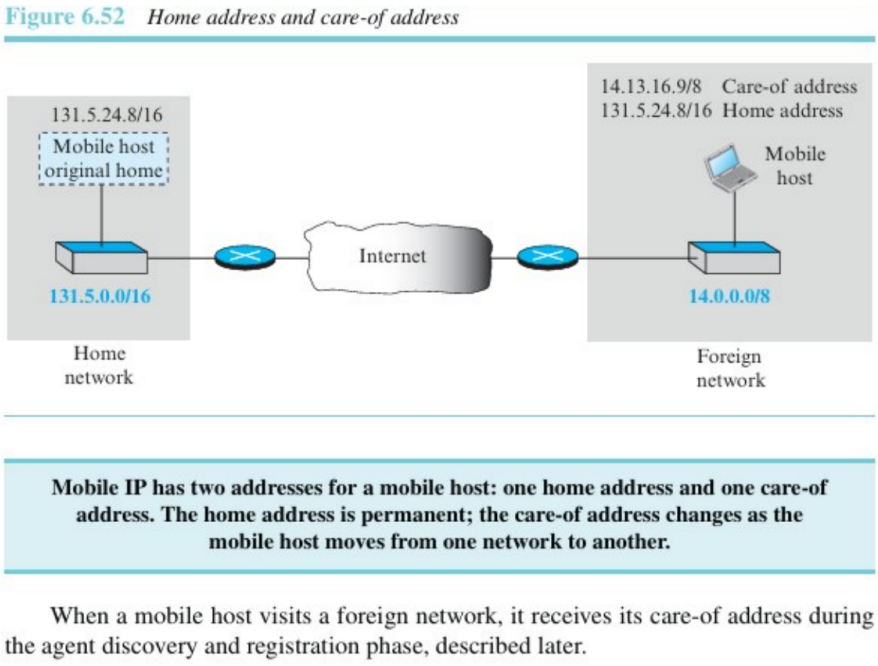


Figure 11. A mobile host with a home address and a care-of address (course module, slide 194, Figure 6.52).

E. Working of Mobile IP

A mobile host goes through three phases to communicate with a remote host (Figure 12).

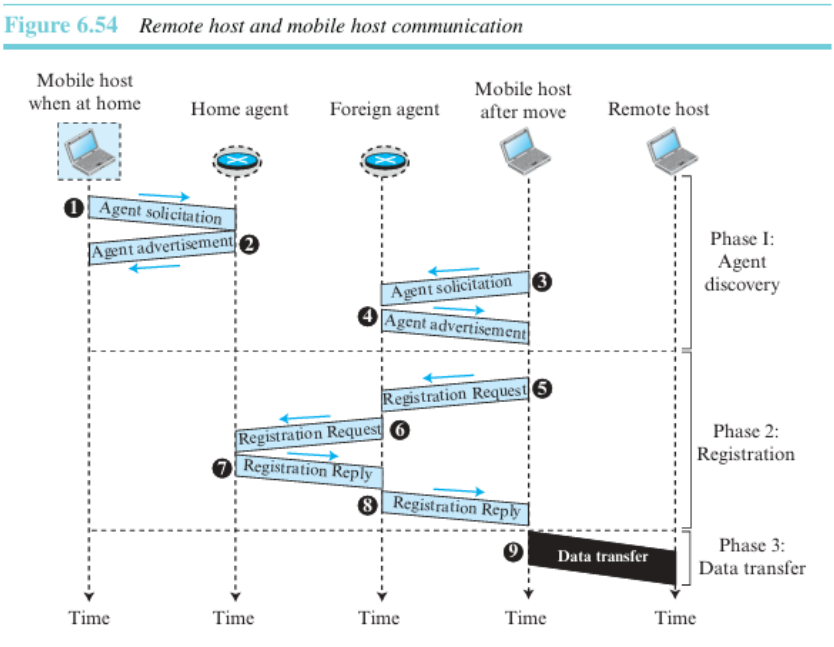


Figure 12. The three phases of Mobile IP communication: agent discovery, registration and data transfer (course module, slide 197, Figure 6.54).

1. **Agent discovery:** The mobile host finds its home agent before leaving home. After it moves, it discovers a foreign agent through agent advertisements (piggybacked on ICMP router advertisements) or sends an agent solicitation if none arrives. It learns the care-of address from the advertisement.

2. **Registration:** The mobile host sends a registration request containing the care-of address, its home address and the home agent address. The foreign agent forwards it to the home agent, which replies with a registration reply through the foreign agent. Registration must be renewed when it expires and cancelled when the host returns home.
3. **Data transfer, remote host to mobile host:** The remote host sends the packet to the home address. The home agent intercepts the packet (using proxy ARP on the home network) and **tunnels** it to the care-of address by encapsulating it in a new IP packet whose destination is the foreign agent. The foreign agent removes the outer header and delivers the original packet to the mobile host.
4. **Data transfer, mobile host to remote host:** The mobile host replies normally, using its home address as the source. The remote host sees packets that appear to come from the home address, so the movement is transparent to it.

F. Routing Inefficiency

Mobile IP can make routes longer than necessary. **Triangle routing** (dog-leg routing) occurs when packets from the remote host always pass through the home agent before reaching the mobile host (Figure 13). Packets from the mobile host to the remote host are not affected. **Double crossing** is more severe: when the mobile host moves to the same site as the remote host, packets from the remote host still cross the internet twice, first to the home agent and then back to that site. One solution to triangle routing is for the home agent to send a binding update to the remote host, so that it can store the care-of address and send future packets directly. The cached address must be updated when the mobile host moves again.

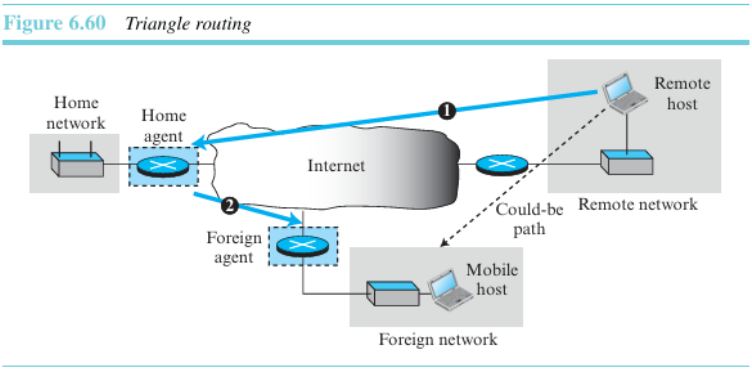


Figure 13. Triangle routing: packets from the remote host pass through the home agent (course module, slide 210, Figure 6.60).

10. Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Aim: To compare the frame structure of wired Ethernet (IEEE 802.3) and wireless LAN (IEEE 802.11) using packet captures in Wireshark, or documented examples where captures are not available.

A. Procedure

1. Open Wireshark, select the wired Ethernet interface and capture while generating traffic (for example, a web page or ping).

2. Apply the display filter **eth** to view Ethernet II frames and expand the Ethernet II header.
3. Select the Wi-Fi interface, capture again, and apply the display filter **wlan** to view IEEE 802.11 frames.
4. Expand the IEEE 802.11 header and compare its fields with those of the Ethernet II header.

Note: a Wi-Fi capture in monitor mode also shows a radiotap header before the 802.11 header. That header is part of the capture tool and is not part of the 802.11 frame.

B. Frame Format Diagrams

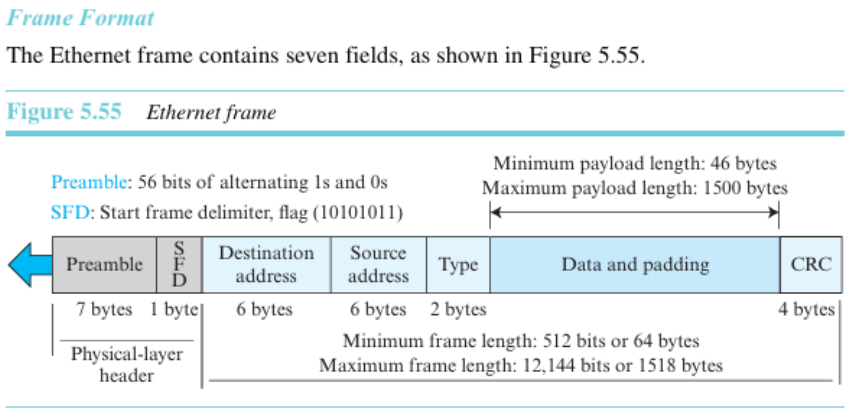


Figure 14. Wired Ethernet (IEEE 802.3) frame (course module, slide 118, Figure 5.55).

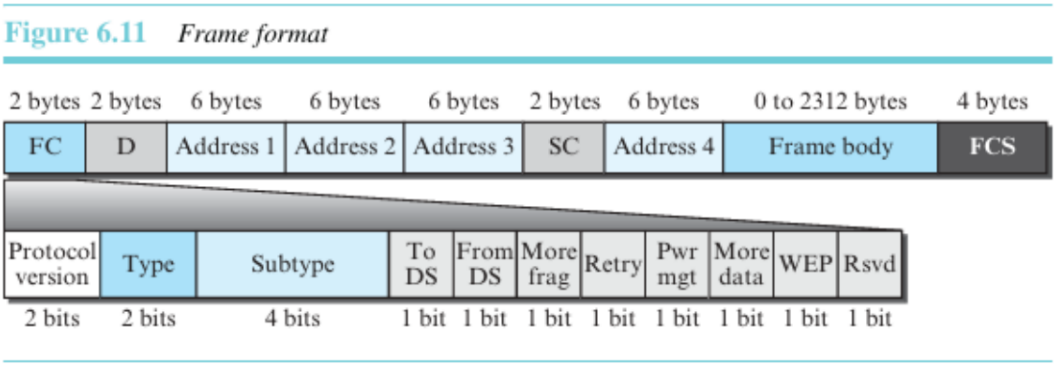


Figure 15. Wireless LAN (IEEE 802.11) frame format (course module, slide 166, Figure 6.11).

The 802.11 frame has two extra fields compared with Ethernet: the Frame Control field, which gives the frame type and flags such as To DS and From DS, and the Duration/ID and Sequence Control fields. Its Address 4 field is used only when a frame travels between two access points in a distribution system.

C. Comparison Table

Feature	Wired Ethernet (IEEE 802.3)	Wireless LAN (IEEE 802.11)
Standard	IEEE 802.3	IEEE 802.11
Medium	Wired cable	Radio waves (air)
Header size	14 bytes (DA 6, SA 6, Type/Length 2)	24 bytes for a typical data frame (FC 2, D 2, three addresses 18, SC 2); 30 bytes when Address 4 is present
Addresses	Two: source and destination MAC	Three or four, and their meaning depends on the To DS and From DS bits (for example BSSID, transmitter, receiver)
Control fields	None beyond Type/Length	Frame Control (type, subtype, To DS, From DS, Retry, Power Management, More Data, WEP), Duration/ID, Sequence Control
Payload (body)	46 to 1500 bytes	0 to 2312 bytes
Error checking	FCS (CRC-32), 4 bytes	FCS (CRC-32), 4 bytes
Overhead in total	18 bytes (14 header + 4 FCS)	28 bytes minimum (24 header + 4 FCS)
Access method	CSMA/CD (classic) or full-duplex switching	CSMA/CA (DCF) with ACK
Common Wireshark filter	eth	wlan

D. Sample Capture Observations

Note on the data: *The table below shows* ***documented sample values****, not the output of a real capture made for this report. They show the field layout a student would look for in Wireshark. Replace them with values from your own captures and attach the screenshots.*

Field	Ethernet example (sample)	Wi-Fi example (sample)
Source address	00:11:22:33:44:55 (SA)	00:1E:C9:5A:7B:2D (Address 2, transmitter)
Destination address	66:77:88:99:AA:BB (DA)	3C:52:82:A1:0F:4B (Address 1, receiver)
Header identification	Ethernet II	IEEE 802.11 data frame
Extra information	EtherType 0x0800 (IPv4)	Frame type/subtype, flags and sequence number
Error check	FCS (CRC-32) at the end of the frame	FCS (CRC-32) at the end of the frame

E. Result

Wired Ethernet frames have a simple header with two MAC addresses and a Type/Length field, whereas 802.11 frames carry three or four addresses and control fields (Frame Control, Duration/ID and Sequence Control) that manage wireless communication. Both use a 4-byte Frame Check Sequence (CRC-32) to detect errors, and in both cases a corrupted frame is discarded rather than corrected.